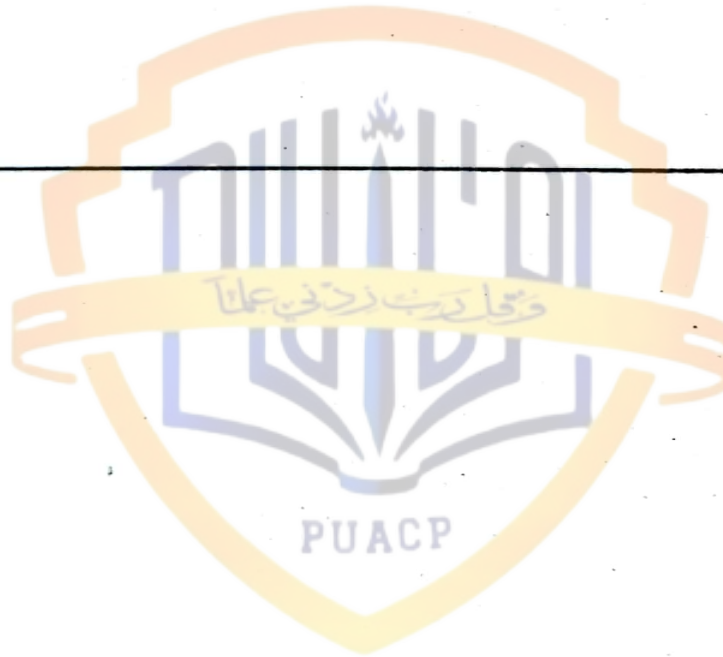


Chapter Overview

- 11.1 Built-in Functions
- 11.2 String Functions
- 11.3 Date & Time Functions
- 11.4 Numeric Functions
- 11.5 Informational Functions
- 11.6 Conversion Functions

Programming Exercise**Short Questions****Multiple Choices****Fill in the Blanks****True/False**

11.1 Built-in Functions

Visual Basic provides many built-in functions for different purposes. These functions save a lot of time. The use of these functions makes Visual Basic programming easier and faster. There are different types of built-in functions:

- String Functions
- Numeric Functions
- Conversion Functions
- Date & Time Functions
- Informational Functions

11.2 String Functions

Strings are used frequently in Visual Basic application. Visual Basic provides many built-in functions to apply different function on strings.

UCase

UCase function is used to convert a string to upper case letters.

```
Name1 = "Usman Khalil"
Name2 = UCase(Name1)
```

The value of Name2 variable will be "USMAN KHALIL".

LCase

LCase function is used to convert a string to lower case letters..

```
Name1 = "Usman Khalil"
Name2 = LCase(Name1)
```

The value of Name2 variable will be "usman khalil".

LTrim

LTrim function is used to remove any space from the left side of a string.

```
Name1 = " Usman Khalil"
Name2 = LTrim(Name1)
```

The value of Name2 variable will be "Usman Khalil".

RTrim

RTrim function is used to remove any space from the right side of a string.

```
Name1 = "Usman Khalil "
```

```
Name2 = RTrim(Name1)
```

The value of Name2 variable will be "Usman Khalil".

Trim

Trim function is used to remove any space from both sides of a string.

```
Name1 = " Usman Khalil "
```

```
Name2 = Trim(Name1)
```

The value of Name2 variable will be "Usman Khalil".

String

String function is used to display or return specified number of character.

```
MsgBox String(10, "+")
```

The above line will display 10 "+" symbols in a message box.

Space

Space function is used to display or return specified number of spaces.

MsgBox "Hello" & Space(10) & "World"

The above line will display "Hello" then 10 spaces and then "World".

StrComp

StrComp function is used to compare two string values and returns an integer. The possible returned values are as follows:

Returned value	Description
1	S1 > S2
0	S1 = S2
-1	S1 < S2

S1 = "Usman"

S2 = "Khalil"

MsgBox StrComp(S1, S2)

The above line will display 1 because "Usman" is greater than "Khalil". The letters that come first in alphabetic are smaller than the letters that come later in alphabetic. For example "A" is smaller than "B". But capital letters are always greater than lower case letter. For example "A" is greater than "b".

Len

Len function is used to determine the total length of a string.

S = "Hello World"

MsgBox Len(S)

The above line will display 11 in a message box.

Left

Left function is used to extract specified number of characters from the left side of a string.

S = "Hello World"

MsgBox Left(S, 2)

The above line will display "He" in a message box.

Right

Right function is used to extract specified number of characters from the right side of a string.

S = "Hello World"

MsgBox Right(S, 3)

The above line will display "rld" in a message box.

Mid

Mid function is used to extract specified number of characters from any part of string.

S = "Hello World"

MsgBox Mid(S, 7, 5)

The above line will display "World" in a message box.

InStr

InStr function is used to search a string for a particular character(s). It returns the position of first occurrence of given character in string. It returns 0 if character is not found.

S = "Hello World"

MsgBox InStr(S, "e")

The above line will display 2 in a message box.

Example

1. Start a new Standard EXE project.
2. Design the form as follows:

3. Add the following code in Click event of Lower Case button.

```
Project1 - Form1 (Code)
cmdLower Click
Private Sub cmdLower_Click()
    Text1.Text = LCase(Text1.Text)
End Sub
```

4. Add the following code in Click event of Upper Case button.

```
Project1 - Form1 (Code)
cmdUpper Click
Private Sub cmdUpper_Click()
    Text1.Text = UCase(Text1.Text)
End Sub
```

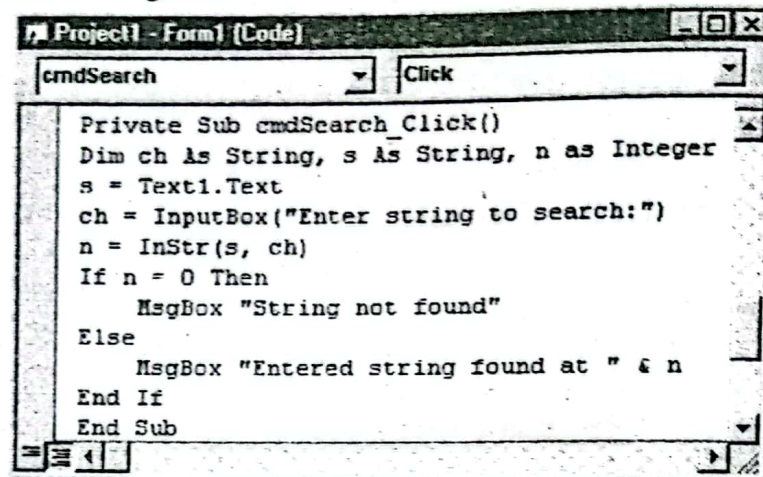
5. Add the following code in Click event of Length button.

```
Project1 - Form1 (Code)
cmdLength Click
Private Sub cmdLength_Click()
    MsgBox "Length of 1st string is " & Len(Text1.Text)
    MsgBox "Length of 2nd string is " & Len(Text2.Text)
End Sub
```

6. Add the following code in Click event of Close button.

```
Project1 - Form1 (Code)
cmdClose Click
Private Sub cmdClose_Click()
    End
End Sub
```

7. Add the following code in Click event of Search button.

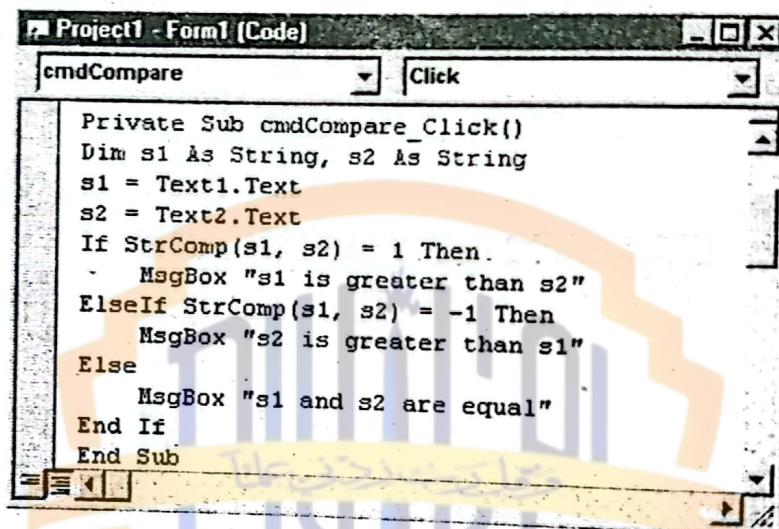


```

Project1 - Form1 (Code)
cmdSearch Click
Private Sub cmdSearch_Click()
    Dim ch As String, s As String, n As Integer
    s = Text1.Text
    ch = InputBox("Enter string to search:")
    n = InStr(s, ch)
    If n = 0 Then
        MsgBox "String not found"
    Else
        MsgBox "Entered string found at " & n
    End If
End Sub

```

8. Add the following code in Click event of Compare button.



```

Project1 - Form1 (Code)
cmdCompare Click
Private Sub cmdCompare_Click()
    Dim s1 As String, s2 As String
    s1 = Text1.Text
    s2 = Text2.Text
    If StrComp(s1, s2) = 1 Then
        MsgBox "s1 is greater than s2"
    ElseIf StrComp(s1, s2) = -1 Then
        MsgBox "s2 is greater than s1"
    Else
        MsgBox "s1 and s2 are equal"
    End If
End Sub

```

9. Run the project.
10. Click different buttons to manipulate the text entered in two text boxes.

11.3 Date & Time Functions

Visual Basic provides different built-in function to manipulate dates and times.

Date

Date function is used to display current system date. The following example will display current date on form in "mm/dd/yyyy" format.

Print Date

Time

Time function is used to display current system time. The following example will display the current time on the form.

Print Time

Now

Now function is used to display both current date and time. The following example will display the current date and time on the form.

Print Now

Day

Day function returns the day part of the date. It uses a parameter of type date. The following example will display 22 on the form.

```
D = "01/22/2003"  
Print Day(D)
```

Month

Month function returns the month part of the date. It uses a parameter of type date. The following example will display 1 on the form.

```
D = "01/22/2003"  
Print Month(D)
```

Year

Year function returns the year part of the date. It uses a parameter of type date. The following example will display 2003 on the form.

```
D = "01/22/2003"  
Print Year(D)
```

Hour

Hour function returns the Hour part of the time. It accepts a parameter of type time. The following example will display 19 on the form. If you change "PM" to "AM" then it will display 7. You can also use "Now" as parameter.

```
T = "7:30:45 PM"  
Print Hour(T)
```

Minute

Minute function returns the Minute part of the time. It accepts a parameter of type time. The following example will display 30 on the form. You can also use "Now" as parameter.

```
T = "7:30:45 PM"  
Print Minute(T)
```

Second

Second function returns the Second part of the time. It accepts a parameter of type time. The following example will display 45 on the form. You can also use "Now" as parameter.

```
T = "7:30:45 PM"  
Print Second(T)
```

WeekDay

WeekDay function returns the day number of the week. The number starts from 1 indicating Sunday. It uses a parameter of type date. The following example will display 4 on the form. The number 4 indicates 4th day i.e. Wednesday.

```
D = "01/22/2003"  
Print WeekKey(D)
```

WeekDayName

WeekDayName function returns the name for the day of the week. The first parameter is an integer representing the day of the week. The second parameter determines if you want

an abbreviated weekday name. A value of **True** will abbreviate the name. The following example will display "Wednesday" on the form:

D = "01/22/2003"

Print WeekDayName(WeekDay(D), False)

The following example will display "Wed" on the form.

Print WeekDayName(WeekDay(D), True)

DateAdd

DateAdd function is used to add or subtract from a date. It has three parameters. The first parameter indicates the part of date to be added or subtracted. The second parameter indicates the amount to be added or subtracted. The third parameter is any valid date to be processed. The following example will display "1/22/2008" on form. It adds 5 years in date stored in D. A negative number is used to subtract an amount from the date.

D = "01/22/2003"

Print DateAdd("yyyy", 5, D)

Option	Description
yyyy	Add or subtract years
m	Add or subtract months
d	Add or subtract days
h	Add or subtract hours
n	Add or subtract minutes
s	Add or subtract seconds

Tables: Options for DateAdd function

DateDiff

DateDiff function is used to find the difference between two dates. It has three parameters. The first parameter indicates the format for displaying the difference. The second and third parameters are any valid dates on which the process will be applied. The following example will display 1 on the form as the difference between two dates is of 1 year.

D1 = "01/22/2003"

D2 = "01/22/2004"

Print DateDiff("yyyy", D1, D2)

Following example will display 365 as the difference between two dates is of 365 days.

Print DateDiff("d", D1, D2)

11.4 Numeric Functions

Numeric functions work on numbers. These functions are used frequently to perform different calculations on numbers.. Following are some important numeric functions.

Abs

Abs function is used to find the absolute value of a number. The absolute value of both -87 and 87 is 87. The following line will display 5:

MsgBox Abs(-5)

Cos

Cos function is used to determine the cosine value of a number.

Log

Log function is used to determine the natural logarithm of a number.

Sin

Sin function is used to determine the sine value of a number.

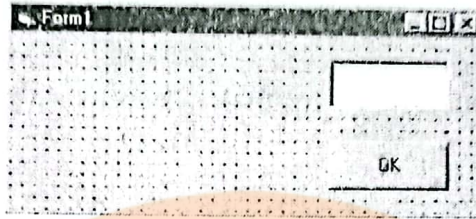
Tan

Tan function is used to determine the tangent value of a number.

Example

The following example explains the use of numeric functions.

1. Start a new Standard EXE project.
2. Design the form as follows:

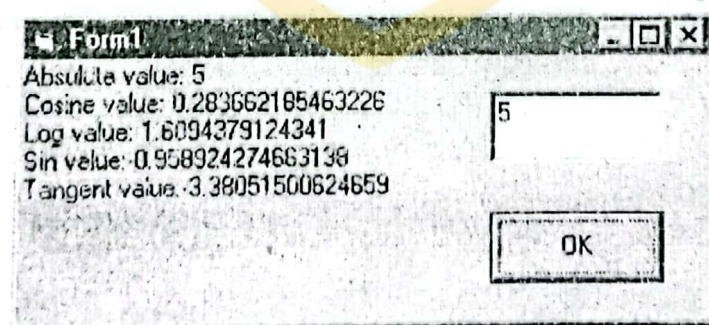


3. Add the following code in the Click event of the button.

```

Project1 - Form1 (Code)
Command1 Click
Private Sub Command1_Click()
    Dim n As Integer
    n = Text1.Text
    Print "Absolute value: " & Abs(n)
    Print "Cosine value: " & Cos(n)
    Print "Log value: " & Log(n)
    Print "Sin value: " & Sin(n)
    Print "Tangent value: " & Tan(n)
End Sub
  
```

4. Run the project.
5. Enter a number in textbox and click OK. The result will appear as follows:

**Round**

Round function is used to round a numeric value with decimal points to the specified digits. The following line will display 5.333

```
MsgBox Round(5.33333, 3)
```

The following line will display 5.667

```
MsgBox Round(5.66666, 3)
```


Rnd

Rnd function is used to generate a random number that is greater than 0 and less than 1. The following line will display a random number between 0 and 1.

```
MsgBox Rnd
```

The above statement is not very useful in this form. It is used with other statements. The following line will display a whole number between 1 and 100.

```
MsgBox Int(Rnd * 100) + 1
```

Sqr

Sqr function is used to find the square root of a number. The following line will display 10.

```
MsgBox Sqr(100)
```

Randomize

Rnd function may generate the same random number again and again. **Randomize** statement is used before **Rnd** function to get a different random each time. The following example will display a whole number between 1 and 100. This time the random numbers will be different each time.

```
Randomize
```

```
MsgBox Int(Rnd * 100) + 1
```

11.5 Informational Functions

Visual Basic provides different informational functions as follows:

VarType

VarType function is used to find the type of a variable. It returns an integer value that indicates a particular type of variable. The following example will display 2 that indicates that the variable is of Integer type.

```
S = 10
```

```
MsgBox VarType(S)
```

The following table shows all numbers for different type of variables.

Value	Meaning	Value	Meaning
0	Empty: Variable not initialized	7	Date
1	Null	8	String
2	Integer	9	OLE object
3	Long Integer	10	Error
4	Single	11	Boolean
5	Double	12	Variant
6	Currency	17	Byte

TypeName

TypeName function is used to find the type name of a variable. It returns a string value that indicates a particular type of variable. The following example will display **Integer** that indicates that the variable is of Integer type:

```
S = 10
```

```
MsgBox TypeName(S)
```

IsNumeric

IsNumeric function is used to check whether the parameter is a numeric value or not. It returns **True** if the value of the parameter is numeric or can be converted to numeric form. The following example will display **True** that indicates that the value is numeric.

```
S = 10  
MsgBox IsNumeric(S)
```

IsDate

IsDate function is used to check whether the parameter is a date/time value or not. It returns **True** if the value of the parameter is date/time or can be converted to date/time form. The following example will return **True** because it can be converted to date value.

```
S = "1/1/2000"  
MsgBox IsDate(S)
```

IsArray

IsArray function is used to check whether the parameter is an array or not. It returns **True** if the parameter is array. The following example will display **False** that indicates that the value is not an array.

```
S = 10  
MsgBox IsArray(S)
```

IsEmpty

IsEmpty function is used to check whether the parameter is empty or not. A variable is empty when it is created but not initialized. It returns **True** if the parameter is empty. The following example will display **True** that indicates that the variable is empty.

```
MsgBox IsEmpty(n)
```

IsNull

IsNull function is used to check whether the parameter is **Null** or not. It returns **True** if the parameter is null. **Null** is normally used in databases. The following example will display **False** that indicates that the value is null.

```
S = 10  
MsgBox IsNull(S)
```

11.6 Conversion Functions

Visual Basic provides different built-in function to convert different values.

Int

Int function is used to drop fractional part of positive number. The following example will display 5.

```
MsgBox Int(5.5)
```

Fix

Fix function is used to drop fractional part of negative number. The following example will display -5.

```
MsgBox Fix(-5.5)
```


CInt

CInt function is used to round a numeric value with decimal points to the nearest whole number. The following example display 9:

MsgBox CInt(8.9)

The following example will display 8:

MsgBox CInt(8.1)

The difference between Int, Fix and CInt functions is as follows:

Original Number	Int	Fix	CInt
3.14	3	3	3
4.7	4	4	5
-1	-1	-1	-1
-3.7	-3	-4	-4
-4.2	-4	-5	-4
4.5	4	4	4
1.5	1	1	2

Val

Val takes a numeric string as parameter and converts it to appropriate numeric data to be used in calculations. If S1 = "1" and S2 = "2", the following example will display 12 not 3.

MsgBox S1 + S2

The following example will display 3:

MsgBox Val(S1) + Val(S2)

CBool

CBool function is used to convert an expression into a **Boolean** value. It returns false if the value is 0 and true for all other numbers. The following example will display true:

MsgBox CBool(5)

CByte

CByte converts an expression to **Byte**. It returns byte value of number from 0 to 255. It gives error for any other number. The following example will display 5:

MsgBox CByte(5)

The following example will generate an error:

MsgBox CByte(256)

CDate

CDate function is used to convert an expression into a date or time value. It returns 12/31/1899 if parameter is 1. The following example will display 1/1/1900:

MsgBox CDate(2)

CDbl

CDbl function is used to convert an expression into double number. It gives error if the expression is not a number.

CCur

CCur function is used to convert an expression into a currency value.

CLng

CLng function is used to convert an expression into a long integer number.

CSng

CSng function is used to convert an expression into a floating-point (decimal) number.

CStr

CStr function is used to convert an expression into a string.

Chr

Chr function is used to associate a character with its ASCII character equivalent. It could be used to convert a number to a character. The following example will display a:

MsgBox Chr(97)

Programming Exercise

1. Write a program that gets the name of the user and displays its length.
2. Write a program that displays current date and time in a label control
3. Write a program that inputs your date of birth and displays your age.

Important Note: The solutions of all questions are available at the end of the book.

Short Questions

Q.1. Which value is stored in variable Ans in the following assignment statements?

- A. Ans = Int(61.32)
- B. Ans = Int(-61.32)
- C. Ans = Fix(-61.32)
- D. Ans = Cint(421.51)

Answer:

- | | |
|-------------|-------------|
| A. 61 | B. -62 |
| C. -61 | D. 422 |

Q.2. Which value appears in string variable AList after these assignments?

- A. AList = UCase("AbCdEfG")
- B. AList = LCase("AbCdEfG")

Answer:

- A. "ABCDEFGH"
- B. "abcdefgh"

Q.3. Why is it useful to include the UCase function in a comparison?

The case of characters is important when comparing strings. An uppercase "Y" is not equal to a lowercase "y". A user may type a name or word in uppercase or lowercase or a combination of two. The use of Ucase or Lcase is very useful in such case. It returns the uppercase or lowercase equivalent string.

Multiple Choice

1. Which of the following is NOT an example of a mathematical function in Visual Basic?
a. Round b. sqr c. len d. exp
2. Which statement will return the number of characters in a textbox named txtEmployee?
a. Length(txtEmployee) b. Len
c. Len(txtEmployee.Text) d. Len("txtEmployee")
3. Which of the following is NOT an example of string function in Visual Basic?
a. Ucase b. Lcase c. Rtrim d. Abs
4. The function Mid("I like Visual Basic", 6) will return:
a. I like b. e
c. e visual Basic d. It is impossible to determine from the information given
5. To remove only leading spaces from a string, use:
a. LTrim() function b. RTrim() function
c. Trim() function d. TrimBoth() function
6. The Val function is used to convert:
a. A number typed into a text box to a numeric value
b. Text typed into a text box to numeric value
c. Both a and b
d. None
7. Which statement is required to make 'Rnd' function generate a different set of random numbers each time the program is executed?
a. Rnd Initialize
b. Rnd = Time
c. Rnd function automatically generates a different set of values each time a program runs
d. Randomize
8. Which of the following function is used to check the parameter is empty or not?
a. isArray b. IsEmpty c. IsNumeric d. None
9. What is the return value of the following code?
LCase("ITSeries")
A. ItSeries B. ITSERIES C. itseries D. Both a and c
10. Which of the following function is used to convert a numeric value to a String?
a. Str b. Val c. CStr d. Len
11. Which of the following functions is used to validate numeric input in a text box?
a. IsValue b. IsNumber c. IsNumeric d. None

Answers

1. c	2. c	3. d	4. c
5. a	6. a	7. d	8. b
9. c	10. a	11. c	

Fill in the Blanks

1. CLng, CCur, and CDbl are all examples of _____ functions.
2. To convert a string type to a numeric value, use the _____ function.
3. The _____ function returns the ASCII code for an individual character.
4. The _____ function returns the character represented by an ASCII code.
5. The _____ function returns a random number greater than or equal to 0, but less than 1.
6. The _____ function returns the current date as set in the computer.
7. The function Len("I Like Visual Basic") returns _____.
8. The _____ function converts a string to all lower-case letters.
9. _____ function is used to remove any space from the left side of a string.
10. _____ function is used to remove any space from the right side of a string.
11. _____ function is used to remove any space both sides of a string.
12. _____ function is used to search a string for a particular characters.
13. _____ function is used to check whether the parameter is an array or not.
14. _____ function is used to round a numeric value with decimal points to the nearest whole number.

Answers

1. Conversion	2. Val	3. Asc
4. Chr	5. Rnd	6. Today()
7. 19	8. Lcase()	9. Ltrim
10. Rtrim	11. Trim	12. InStr
13. IsArray	14. CInt	15.

True/False

1. The CDbl intrinsic function will take a double data type and convert it into another data type.
2. The IsNumeric intrinsic function will return a value of true if the string can be changed into a numeric value.
3. the Int function is used to drop the fractional part of a positive number.
4. IsNumeric function is used to check whether the parameter is Null or not.
5. The Sqrt function is used to find the square root of a number.
6. Right function is used to extract specified number of characters from any part of a string.
7. The Len function is used to determine the length of a list variable (array).
8. WeekdayName is an example of a "date and time" function provided in Visual Basic.

Answers

1. F	2. T	3. T	4. F
5. F	6. F	7. F	8. T

Project 7

Scientific Calculator

Project Specification

The project Scientific Calculator displays a scientific calculator for calculations.

Procedure

1. Start a new Standard EXE project.
2. Place the following controls on the form with the given properties:

Control	Control Name	Property	Value
Label	Result	Caption	0
		BackColor	White
		Alignment	1 – Right Justify
		BorderStyle	1 – Fixed Single
CommandButton	Num(0)	Caption	0
CommandButton	Num(1)	Caption	1
CommandButton	Num(2)	Caption	2
CommandButton	Num(3)	Caption	3
CommandButton	Num(4)	Caption	4
CommandButton	Num(5)	Caption	5
CommandButton	Num(6)	Caption	6
CommandButton	Num(7)	Caption	7
CommandButton	Num(8)	Caption	8
CommandButton	Num(9)	Caption	9
CommandButton	Sqrt	Caption	Sqrt
CommandButton	Backspace	Caption	Backspace
CommandButton	C	Caption	C
CommandButton	Exit	Caption	Exit
CommandButton	Log	Caption	Log
CommandButton	Sin	Caption	Sin
CommandButton	Cos	Caption	Cos
CommandButton	Tan	Caption	Tan
CommandButton	Equal	Caption	=
CommandButton	Point	Caption	.
CommandButton	Percent	Caption	%
CommandButton	X1	Caption	1/x
CommandButton	X2	Caption	X^2
CommandButton	X3	Caption	X^3

Note: The project uses two controls arrays of buttons:

- One control array is used for buttons displaying numbers from 0 to 9. The indexes of the buttons in the array are same as their captions.

- Second control is used for buttons displaying mathematical operators. The index of "/" button is 0, index of "*" button is 1, index of "-" button is 2 and index of "+" button is 3. The remaining buttons are individual buttons.

3. Design the form as follows:

IT Series Calculator					
0					
Sqrt	Backspace	C	Exit		
Log	7	8	9	/	%
Sin	4	5	6	*	1/x
Cos	1	2	3	-	x^2
Tan	0	=	.	+	x^3

4. Add the following code in General Declaration section:

```

Project1 - Form1 (Code)
(General) (Declarations)
Dim N1 As Double, N2 As Double, Opr As Integer
  
```

5. Add the following code in Click event of Num control array:

```

Project1 - Form1 (Code)
Num Click
Private Sub Num_Click(Index As Integer)
    If Result = "0" Then
        Result = Index
    Else
        Result = Result & Index
    End If
End Sub
  
```

6. Add the following code in Click event of Op control array:

```

Project1 - Form1 (Code)
Op Click
Private Sub Op_Click(Index As Integer)
    N1 = Result
    Opr = Index
    Result = ""
End Sub
  
```


7. Add the following code in Click event of Sqrt button.

```

Project1 - Form1 (Code)
Sqrt Click
Private Sub Sqrt_Click()
    If Result < 0 Then
        MsgBox "Cannot calculate Sqrt of negative value"
    Else
        Result = Sqr(Result)
    End If
End Sub

```

8. Add the following code in Click event of Log button.

```

Project1 - Form1 (Code)
Log Click
Private Sub Log_Click()
    If Result <= 0 Then
        MsgBox ("Cannot find log of this number")
    Else
        Result = Math.Log(CDbl(Result))
    End If
End Sub

```

9. Add the following code in Click event of Sin button.

```

Project1 - Form1 (Code)
Sin Click
Private Sub Sin_Click()
    Result = Math.Sin(CDbl(Result))
End Sub

```

10. Add the following code in Click event of Cos button.

```

Project1 - Form1 (Code)
Cos Click
Private Sub Cos_Click()
    Result = Math.Cos(CDbl(Result))
End Sub

```

11. Add the following code in Click event of Tan button.

```

Project1 - Form1 (Code)
Tan Click
Private Sub Tan_Click()
    Result = Math.Sin(CDbl(Result))
End Sub

```

12. Add the following code in Click event of Backspace button.

```

Project1 - Form1 (Code)
Backspace Click
Private Sub Backspace_Click()
    Result = Mid(Result, 1, (Len(Result) - 1))
    If Result = "" Then Result = "0"
End Sub

```

13. Add the following code in Click event of C button.

```
Project1 - Form1 (Code)
C Click
Private Sub C_Click()
    Result = "0"
    N1 = 0
    N2 = 0
End Sub
```

14. Add the following code in Click event of Exit button.

```
Project1 - Form1 (Code)
Exit Click
Private Sub Exit_Click()
    If MsgBox("Exit?", vbYesNo, "Exit") = vbYes Then
        End
    End If
End Sub
```

15. Add the following code in Click event of Percent button.

```
Project1 - Form1 (Code)
Percent Click
Private Sub Percent_Click()
    Result = N1 * CDbl(Result) / 100
End Sub
```

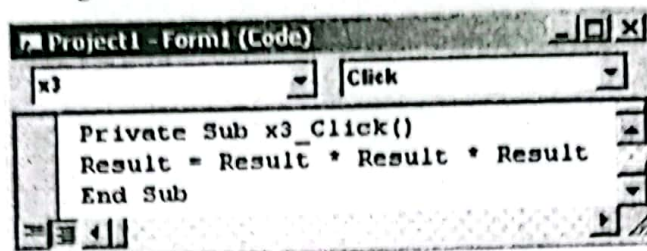
16. Add the following code in Click event of x1 button.

```
Project1 - Form1 (Code)
x1 Click
Private Sub x1_Click()
    If Result = 0 Then
        MsgBox "Cannot find 1/x!", vbExclamation
    Else
        Result = 1 / Result
    End If
End Sub
```

17. Add the following code in Click event of x2 button.

```
Project1 - Form1 (Code)
x2 Click
Private Sub x2_Click()
    Result = Result * Result
End Sub
```

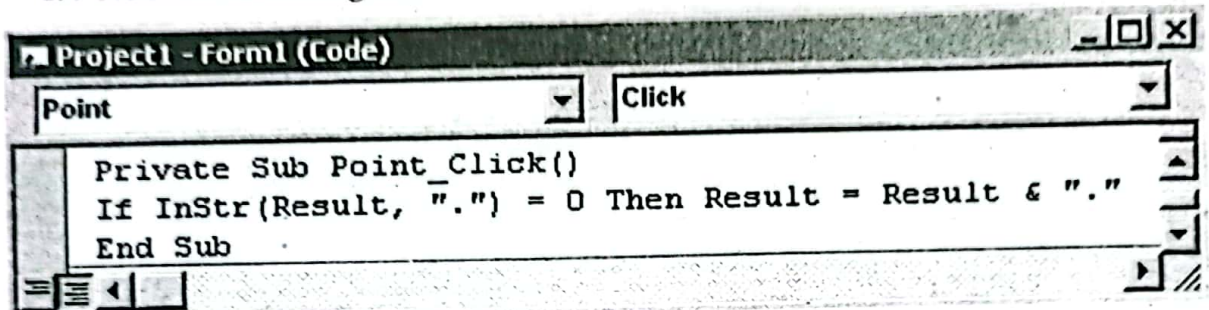

18. Add the following code in Click event of x3 button.



```

Project1 - Form1 (Code)
x3 Click
Private Sub x3_Click()
    Result = Result * Result * Result
End Sub
  
```

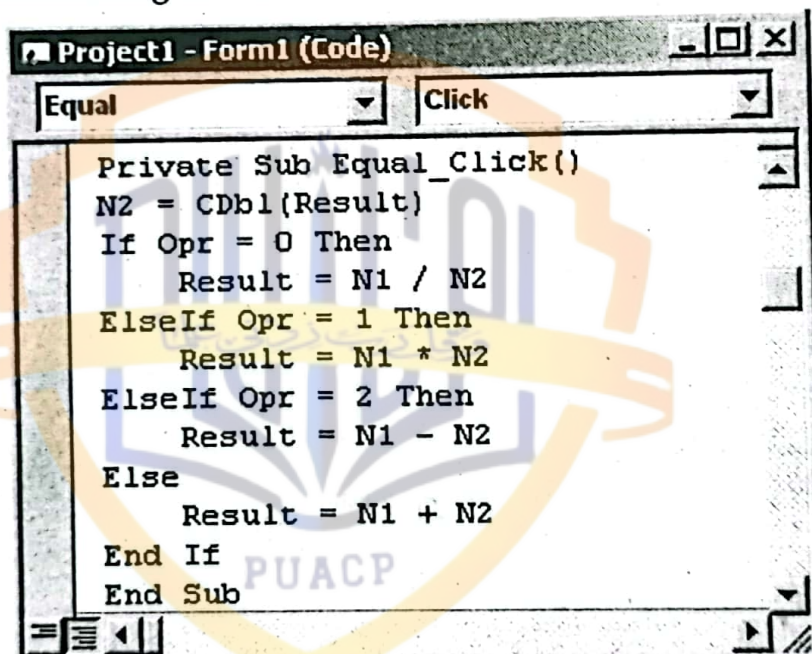
19. Add the following code in Click event of Calculate Bill button.



```

Project1 - Form1 (Code)
Point Click
Private Sub Point_Click()
    If Instr(Result, ".") = 0 Then Result = Result & "."
End Sub
  
```

20. Add the following code in Click event of Calculate Bill button.



```

Project1 - Form1 (Code)
Equal Click
Private Sub Equal_Click()
    N2 = CDb1(Result)
    If Opr = 0 Then
        Result = N1 / N2
    ElseIf Opr = 1 Then
        Result = N1 * N2
    ElseIf Opr = 2 Then
        Result = N1 - N2
    Else
        Result = N1 + N2
    End If
End Sub
  
```

21. Run the project and check the working of calculator.